

Cloudflare Radar 2024 Year in Review

Internet Patterns and Trends Observed Through Cloudflare's Global Network

Data-Driven Analysis for Technology Executives and Network Infrastructure Leaders

Source: radar.cloudflare.com/year-in-review/2024

17% Global Traffic Growth and Starlink's 3.3x Surge

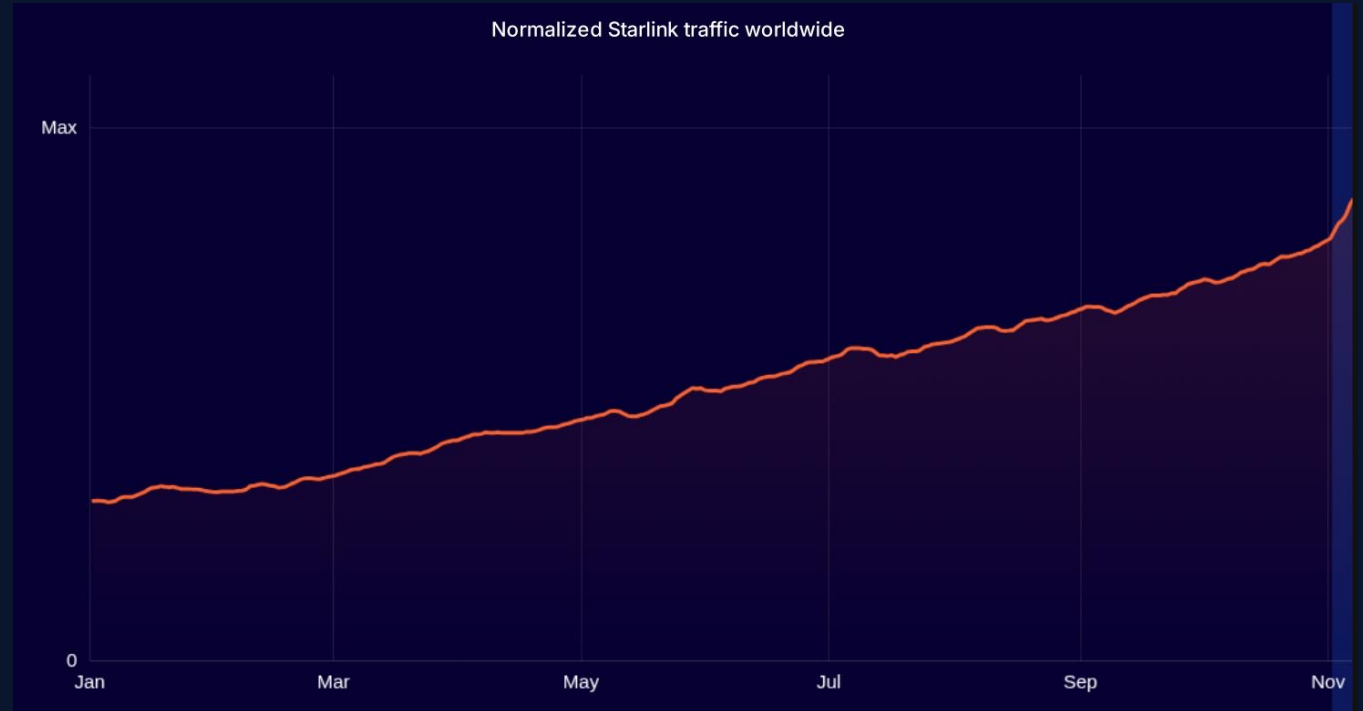
17%

Year-over-year global Internet traffic growth in 2024

3.3x

SpaceX Starlink traffic surge as satellite connectivity expands

- Traffic trend normalized from mid-January to account for post-holiday activity
- Starlink trend scaled relative to peak observed request volume
- Growth driven by communication, commerce, entertainment, and transportation



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Top 10 most popular Internet services worldwide (based on 1.1.1.1 DNS resolver data)

Insight: Generative AI continued its rapid rise in 2024, while the Metaverse receded into the background.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-quantum encryption share of TLS 1.3 traffic in 2024

April 17, 2024

Google Chrome 124 enabled post-quantum key agreement by default

- Mozilla Firefox began rolling out post-quantum by default
- Apple Safari started initial testing of post-quantum encryption
- Surge reflects industry urgency for quantum-resistant security



Post-quantum encrypted traffic worldwide showing rapid adoption after Chrome 124 release

HTTP/3: 21%

of traffic

Faster, encrypted-by-default protocol

Go: #1 API

Client Language

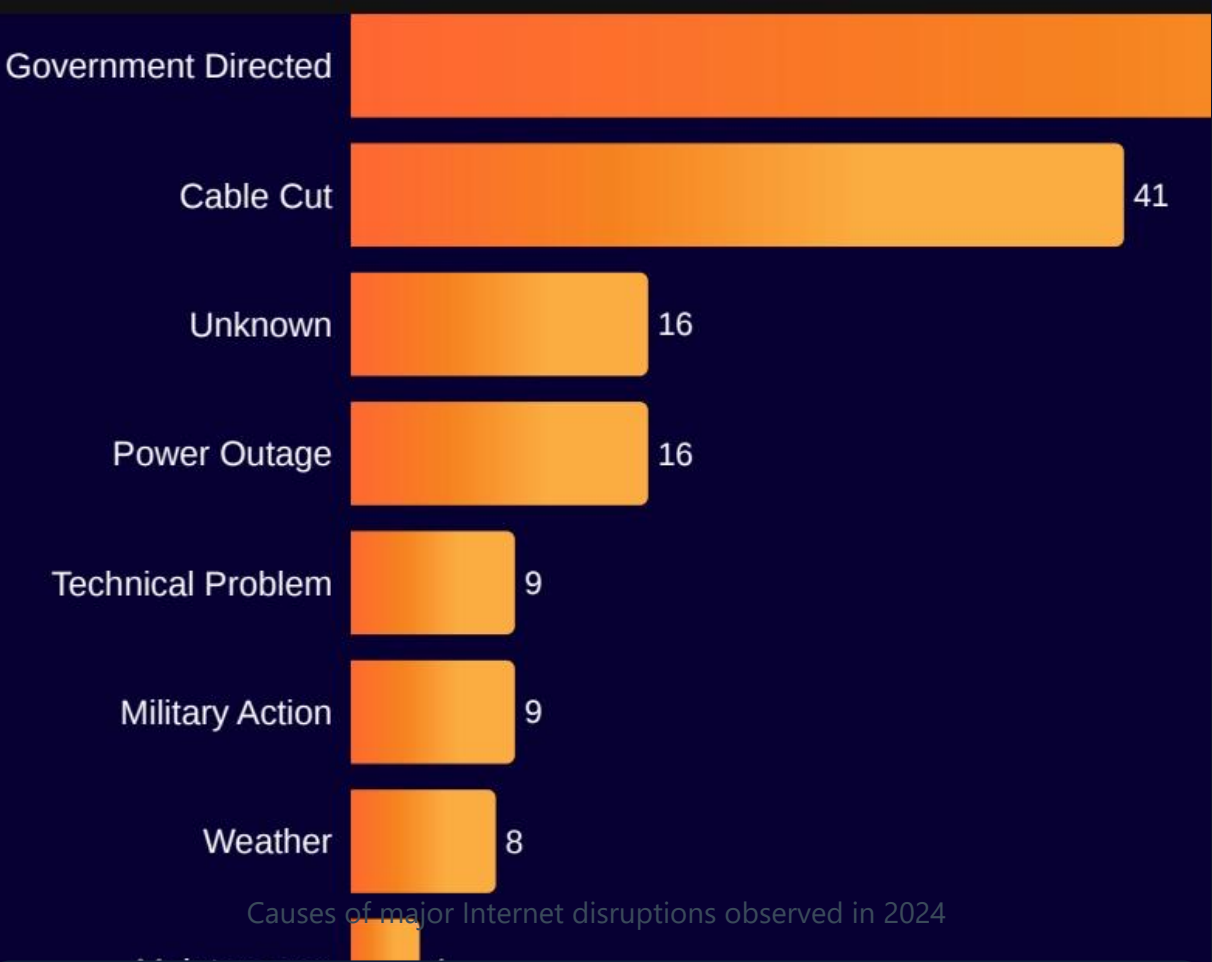
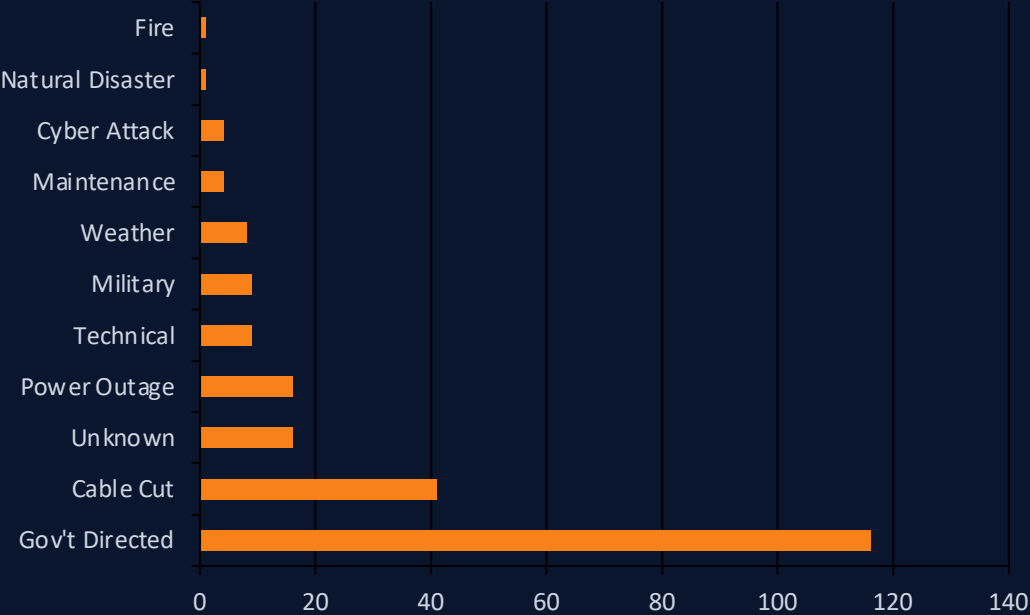
~12% of automated API requests

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet disruptions observed globally in 2024

Disruption Causes (2024)



Government-directed shutdowns (116+) accounted for over half of all major disruptions.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all traffic mitigated by Cloudflare

68%

of global bot traffic from top 10 countries

4.3%

of emails processed were malicious

Most Attacked Sector: Gambling and Games

Log4j (CVE-2021-44228) continues to be actively targeted — more than two years after initial disclosure

RPKI valid routes increased by 6.4%, improving Internet routing security

Email Security Threats

- 43% of malicious emails contained a deceptive link
- .bar TLD originated the largest share of malicious and spam email
- Email remains a primary vector for social engineering attacks

Key Insight: The persistence of Log4j exploitation underscores that widely known vulnerabilities remain active threats years after disclosure — patching alone is insufficient without continuous monitoring.

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY, with Starlink traffic surging 3.3x as satellite connectivity expands to underserved regions.

02

Post-Quantum Encryption Momentum

Adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

03

Connectivity Fragility

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed globally.

04

Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

05

Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing roughly 12% of automated API traffic.

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